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Lab Program-1.

WAP to stimulate the working of stack using an array with the working following -

a) Push b) Pop c) Display.

The program should print appropriate messages for stack overflow, stack underflow.

```
# include <stdio.h>
# define N 5
int stack [N];
int top = -1;
void push () {
    int a;
    printf ("Enter a number: ");
    scanf ("%d", &a);
    if (top == N-1){
        printf ("Overflow");
    }
    else {
        top ++;
        stack [top] = a;
        printf ("The top element : %d", stack[top]);
    }
}

void pop () {
    int item;
    if (top == -1) {
        printf ("Underflow");
    }
    else {
        item = stack [top];
        top --;
        printf ("Popped element : %d", item);
    }
}
```

```
void peek(){
    int item;
    if (top == -1){
        printf(" Underflow");
    }
    else {
        print f("peek element = %d", stack[top]);
    }
}

void main(){
    int stack[5];
    int top = -1;
    int ch;
    do {
        printf("\n Enter 1,2,3 for insertion, deletion and peek:");
        scanf("%d", &ch);
        switch (ch){
            case 1:
                push();
                break;
            case 2:
                pop();
                break;
            case 3:
                peek();
                break;
            default:
                printf("invalid input");
        }
    } while (ch != -1);
}
```

Output:

Enter 1,2,3 for insertion, deletion and peek: 1
Enter a number: 14
The top element: 14.
Enter 1,2,3 for insertion, deletion and peek: 1
Enter a number: 45.
~~Enter 1,2,3~~ The top element: 45
Enter 1,2,3 for insertion, deletion and peek: 2
~~Ent~~ Popped element = 45.
Enter 1,2,3 for insertion, deletion and peek: 3
Peek element = 14.
Enter 1,2,3 for insertion, deletion and peek: -1
invalid input.

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